

# SMACT v4: Enhanced Screening and Prediction Tools for Computational Materials Design

Kinga O. Mastej<sup>1</sup>, Anthony Onwuli<sup>1</sup>, Alexander Moriarty<sup>2</sup>, Daniel W. Davies<sup>1</sup>, Ryan Nduma<sup>1</sup>, Keith T. Butler<sup>4</sup>, Jiwoo Lee<sup>3</sup>, Masahiro Negishi<sup>1</sup>, Panyalak Detrattanawichai<sup>1</sup>, Alex M. Ganose<sup>5</sup>, Hyunsoo Park<sup>1</sup>, Rhys E. A. Goodall<sup>6</sup>, Tianshu Li<sup>1</sup>, and Aron Walsh<sup>1</sup>

<sup>1</sup> Department of Materials, Imperial College London, London, United Kingdom <sup>2</sup> Department of Chemical Engineering, University College London, London, United Kingdom <sup>3</sup> Department of Materials Science and Engineering, Yonsei University, Seoul, South Korea <sup>4</sup> Department of Chemistry, University College London, London, United Kingdom <sup>5</sup> Department of Chemistry, Imperial College London, London, United Kingdom <sup>6</sup> Radical AI, New York, United States of America

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

## Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Editor: [Open Journals](#)

## Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

## License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

## Summary

SMACT (Semiconducting Materials by Analogy and Chemical Theory) is a Python library for the high-throughput screening and discovery of inorganic materials. The name reflects foundational analogy-based approaches to semiconductor prediction (Goodman, 1958; Pamplin, 1964). Originally published as a JOSS paper in 2019 (Davies et al., 2019), the package provided chemical filters based on charge neutrality, electronegativity, and elemental data to rapidly identify chemically plausible compositions (Davies et al., 2016). Since then, SMACT has undergone substantial development from v2.1 through v4.0. The library now includes modules for crystal structure prediction via data-mined ionic substitution (Hautier et al., 2011), dopant prediction, composition-to-property prediction using deep learning models (Goodall & Lee, 2020), and tools for mapping inorganic crystal chemical space (Park, Onwuli, et al., 2025). These additions transform SMACT from a compositional screening tool into a comprehensive platform for low-cost computational materials design.

## Statement of Need

The search space for new inorganic materials is immense, exceeding  $4 \times 10^{12}$  quaternary element combinations (Davies et al., 2016), and continues to grow as multi-component systems attract interest. Researchers need tools that can rapidly filter this space using established chemical principles before committing to expensive first-principles calculations or synthesis. The original SMACT addressed this need through charge neutrality and electronegativity screening. However, the field has since advanced: researchers now require not only compositional screening but also structure prediction, property estimation, and integration with modern machine learning workflows.

SMACT v4 meets these needs by providing a unified Python library where a researcher can (i) enumerate and filter compositions, (ii) predict likely crystal structures from data-mined substitution probabilities, (iii) estimate material properties directly from stoichiometry without requiring structural information, and (iv) identify candidate dopants for known host materials. The stages share a common representation of compositions and oxidation-state-resolved species, so screening output can be passed on to structure, property or dopant prediction as the question requires, and of the four only structure prediction requires structural input, in the form of the parent structures it substitutes into; a campaign can therefore narrow a chemical space before

incurring any structural cost. The library is designed for desktop-scale computation, with data caching and multiprocessing support that enables screening of millions of compositions in minutes.

## State of the Field

Several established libraries serve the computational materials science community. Pymatgen (Ong et al., 2013) provides comprehensive tools for structure manipulation and thermodynamic analysis. The Atomic Simulation Environment (ASE) (Larsen et al., 2017) focuses on atomistic simulations. Matminer (Ward et al., 2018) specialises in featurisation and machine learning pipelines for materials data. These tools primarily operate on known structures or experimental data.

SMACT occupies a distinct niche: it works at the composition level, upstream of structure-based tools, to define which regions of chemical space merit further investigation. This top-down, composition-first approach complements structure-based workflows with lower-cost tools that scale to vast chemical spaces. The structure prediction and property prediction modules added since v2.0 provide natural bridges between SMACT's compositional screening and downstream structural analysis tools. The ElementEmbeddings integration (Antunes et al., 2022) further connects SMACT to modern representation learning approaches for materials informatics (Onwuli et al., 2024).

The distinction is sharpest where the functionality appears to overlap. Pymatgen implements the same data-mined ionic substitution model (Hautier et al., 2011) in its SubstitutionProbability, SubstitutionPredictor, Substitutor and SubstitutionPredictorTransform classes, and predicts dopants through `get_dopants_from_substitution_probabilities` and `get_dopants_from_shannon_radii`. Both of those dopant predictors require a crystal structure decorated with oxidation states, and the radius-based one additionally requires a bonded structure graph, because Shannon radii are defined per coordination number. SMACT's Doper takes a list of species instead, and so answers what could be doped into a *composition* before any structure is known, which is the regime in which the candidate space is largest. The same reasoning shapes SMACT's structure prediction, which pairs the substitution model with a local, species-indexed database of known structures so that candidate compositions are matched against parent structures in bulk rather than one transformation at a time.

SMACT therefore extends the substitution model rather than reimplementing it: substitution tables can be replaced by learned, oxidation-state-aware species embeddings (Onwuli et al., 2024), and dopant candidates on cation sites are ranked by selectivity between competing host sites as well as by raw probability. Both presuppose the structure-free setting, which is why they sit here rather than upstream in pymatgen, on which SMACT continues to depend for structure handling and I/O.

## Software Design

SMACT v4 retains the Element and Species classes at its core, providing access to tabulated data for all elements in arbitrary oxidation states and coordination environments. The following major modules have been added since the original publication.

**Structure prediction.** Implements data-mined ionic substitution methods (Hautier et al., 2011) to predict likely crystal structures for new compositions by analogy with known parent structures, using substitution probabilities derived from statistical analysis of experimental databases.

**Dopant prediction.** Enables high-throughput identification of p-type and n-type dopants for a given host composition. Candidates are filtered on accessible oxidation states, an n-type dopant being required to carry a higher charge than the host ion it replaces and a p-type dopant

a lower one, in both cases retaining the sign of the site, then scored either by data-mined substitution probability (Hautier et al., 2011) or by cosine similarity in a species embedding (Onwuli et al., 2024). For substitutions on cation sites, the default ranking combines that score with a selectivity term weighing a dopant's affinity for one host site against its affinity for the others. The workflow takes a composition rather than a structure, so no ionic-radius criterion is applied: Shannon radii are defined per coordination number, which is not known until a structure is.

**Property prediction.** Provides composition-to-property prediction using pre-trained ROOST models (Goodall & Lee, 2020), allowing users to predict band gaps directly from chemical formulae with uncertainty estimates.

**Oxidation states and metallicity.** Implements a probabilistic model for predicting the likelihood of metal species coexisting in compounds based on anion-dependent statistics (Davies et al., 2018), with updated default data (ICSD24). A metallicity scoring module distinguishes ionic compounds from intermetallic phases.

**Crystal space utilities.** Tools for systematically mapping inorganic composition space (Park, Onwuli, et al., 2025), including data acquisition, composition generation, and dimensionality-reduced visualisation of compositional embeddings.

**Screening enhancements.** The core screening functions now support mixed-valence compositions, switchable oxidation state datasets, and consensus-based filtering.

## Research Impact Statement

SMACT has been employed in major materials discovery campaigns, including GNoME from Google DeepMind (Merchant et al., 2023) and MatterGen from Microsoft Research (Zeni et al., 2025). Park et al. (Park, Onwuli, et al., 2025) used SMACT's screening and crystal space tools to map the landscape of inorganic crystal chemistry. Onwuli et al. (Onwuli et al., 2024) employed SMACT's species representations for materials informatics, developing ionic embeddings that capture oxidation-state-dependent chemical behaviour. Park et al. (Park, Mastej, et al., 2025) used SMACT in a workflow for closing the synthesis gap in computational materials design. Nduma et al. (Nduma et al., 2025) integrated SMACT into the Crystallise multi-tool agent for autonomous materials design.

## AI Usage Disclosure

Generative AI tools (Claude, Anthropic) were used to assist with code modernisation tasks during the v4.0.0 development cycle. All AI-generated code was reviewed and validated by the authors.

## Author Contributions

**KOM** is the current maintainer who contributed consensus-based oxidation state filtering for `smact_validity` and `smact_filter`, and led the v4.0.0 release including mixed-valence support, a deep code audit, full type annotations, security hardening, test migration to pytest with coverage raised to over 85%, and CI consolidation. **AO** was the primary developer from v2.2 to v3.1, contributing the dopant prediction module, oxidation states improvements, crystal space utilities, and extensive maintenance. **AM** designed and implemented the structure prediction subpackage and benchmarking framework. **DWD** is the original lead developer and contributed to CI infrastructure and code quality tooling. **RN** implemented the property prediction subpackage, metallicity module, and screening performance optimisations. **KTB** is an original author who continued contributing to the codebase. **JL** contributed the initial dopant prediction implementation and charge state comparison fixes. **MN** updated and fixed

all example and tutorial notebooks to work correctly with the current SMACT library. PD contributed to lattice parameter calculations. AMG contributed to the implementation of mixed-valence support during the Alchemy's SMACT hackathon. HP contributed the crystal space visualisation tools and element data corrections. REAG fixed a screening bug related to electronegativity handling. TL contributed to code reorganisation and utility refactoring. AW supervised the project and contributed to the codebase.

## Acknowledgements

We acknowledge contributions from all members of the Materials Design Group at Imperial College London who have provided feedback, testing, and feature requests. We thank the JOSS editors and reviewers of the original 2019 paper. This work was supported by multiple funders over the years, including Engineering and Physical Sciences Research Council (UKRI), the Royal Society, the Leverhulme Trust, and the European Research Council.

## References

- Antunes, L. M., Grau-Crespo, R., & Butler, K. T. (2022). Distributed representations of atoms and materials for machine learning. *Npj Computational Materials*, 8, 44. <https://doi.org/10.1038/s41524-022-00729-3>
- Davies, D. W., Butler, K. T., Isayev, O., & Walsh, A. (2018). Materials discovery by chemical analogy: Role of oxidation states in structure prediction. *Faraday Discussions*, 211, 553–568. <https://doi.org/10.1039/C8FD00032H>
- Davies, D. W., Butler, K. T., Jackson, A. J., Morris, A., Frost, J. M., Skelton, J. M., & Walsh, A. (2016). Computational screening of all stoichiometric inorganic materials. *Chem*, 1(4), 617–627. <https://doi.org/10.1016/j.chempr.2016.09.010>
- Davies, D. W., Butler, K. T., Jackson, A. J., Skelton, J. M., Morita, K., & Walsh, A. (2019). SMACT: Semiconducting materials by analogy and chemical theory. *Journal of Open Source Software*, 4(38), 1361. <https://doi.org/10.21105/joss.01361>
- Goodall, R. E. A., & Lee, A. A. (2020). Predicting materials properties without crystal structure: Deep representation learning from stoichiometry. *Nature Communications*, 11, 6280. <https://doi.org/10.1038/s41467-020-19964-7>
- Goodman, C. H. L. (1958). The prediction of semiconducting properties in inorganic compounds. *Journal of Physics and Chemistry of Solids*, 6(4), 305–314. [https://doi.org/10.1016/0022-3697\(58\)90050-7](https://doi.org/10.1016/0022-3697(58)90050-7)
- Hautier, G., Fischer, C., Ehrlacher, V., Jain, A., & Ceder, G. (2011). Data mined ionic substitutions for the discovery of new compounds. *Inorganic Chemistry*, 50(2), 656–663. <https://doi.org/10.1021/ic102031h>
- Larsen, A. H., Mortensen, J. J., Blomqvist, J., Castelli, I. E., Christensen, R., Duřak, M., Friis, J., Groves, M. N., Hammer, B., Hargus, C., Hermes, E. D., Jennings, P. C., Jensen, P. B., Kermode, J., Kitchin, J. R., Kolsbjerg, E. L., Kubal, J., Kaasbjerg, K., Lysgaard, S., ... Jacobsen, K. W. (2017). The atomic simulation environment—a Python library for working with atoms. *Journal of Physics: Condensed Matter*, 29(27), 273002. <https://doi.org/10.1088/1361-648X/aa680e>
- Merchant, A., Batzner, S., Schoenholz, S. S., Aykol, M., Cheon, G., & Cubuk, E. D. (2023). Scaling deep learning for materials discovery. *Nature*, 624, 80–85. <https://doi.org/10.1038/s41586-023-06735-9>
- Nduma, R., Park, H., & Walsh, A. (2025). *Crystallise: A multi-tool agent for materials design*. <https://doi.org/10.48550/arXiv.2512.00977>

- 179 Ong, S. P., Richards, W. D., Jain, A., Hautier, G., Kocher, M., Cholia, S., Gunter, D., Chevrier,  
180 V. L., Persson, K. A., & Ceder, G. (2013). Python materials genomics (pymatgen): A  
181 robust, open-source python library for materials analysis. *Computational Materials Science*,  
182 68, 314–319. <https://doi.org/10.1016/j.commatsci.2012.10.028>
- 183 Onwuli, A., Butler, K. T., & Walsh, A. (2024). Ionic species representations for materials  
184 informatics. *APL Machine Learning*, 2(3), 036112. <https://doi.org/10.1063/5.0227009>
- 185 Pamplin, B. R. (1964). A systematic method of deriving new semiconducting compounds  
186 by structural analogy. *Journal of Physics and Chemistry of Solids*, 25(7), 675–684.  
187 [https://doi.org/10.1016/0022-3697\(64\)90176-3](https://doi.org/10.1016/0022-3697(64)90176-3)
- 188 Park, H., Mastej, K. O., Detrattanawichai, P., Nduma, R., & Walsh, A. (2025). *Closing the*  
189 *synthesis gap in computational materials design*. [https://doi.org/10.26434/chemrxiv-2025-](https://doi.org/10.26434/chemrxiv-2025-sbc0c-v2)  
190 [sbc0c-v2](https://doi.org/10.26434/chemrxiv-2025-sbc0c-v2)
- 191 Park, H., Onwuli, A., Butler, K. T., & Walsh, A. (2025). Mapping inorganic crystal chemical  
192 space. *Faraday Discussions*, 256, 601–613. <https://doi.org/10.1039/D4FD00063C>
- 193 Ward, L., Dunn, A., Faghaninia, A., Zimmermann, N. E. R., Bajaj, S., Wang, Q., Montoya,  
194 J., Chen, J., Bystrom, K., Dylla, M., Chard, K., Asta, M., Persson, K. A., Snyder, G. J.,  
195 Foster, I., & Jain, A. (2018). Matminer: An open source toolkit for materials data mining.  
196 *Computational Materials Science*, 152, 60–69. [https://doi.org/10.1016/j.commatsci.2018.](https://doi.org/10.1016/j.commatsci.2018.05.018)  
197 [05.018](https://doi.org/10.1016/j.commatsci.2018.05.018)
- 198 Zeni, C., Pinsler, R., Zügner, D., Fowler, A., Horton, M., Fu, X., Wang, Z., Shysheya, A.,  
199 Crabbé, J., Ueda, S., Sordillo, R., Sun, L., Smith, J., Nguyen, B., Schulz, H., Lewis, S.,  
200 Huang, C.-W., Lu, Z., Zhou, Y., ... Xie, T. (2025). A generative model for inorganic  
201 materials design. *Nature*, 639, 624–632. <https://doi.org/10.1038/s41586-025-08628-5>